Jingyuan Hou

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Available: Jan. 2026 – Present

Education

Northeastern University Portland, ME
Master of Science in Artificial Intelligence GPA: 3.58/4.0 *Sep. 2025 – May 2027*

Virginia Tech Blacksburg, VA
Bachelor of Science in Computer Science GPA: 3.62/4.0 *Sep. 2021 – May 2025*

Technical Knowledge

Languages: Python, Java, C, C++, Go, JavaScript, TypeScript, Kotlin, R, SQL, Bash/Shell, RISC-V
Frameworks: PyTorch, TensorFlow, Keras, LangChain, LangGraph, React, Next.js, Vue.js, Node.js, Express.js, Flask, FastAPI, Vite, SQLAlchemy, Celery
Data & Tools: PostgreSQL, SQLite, Redis, NumPy, HTML/CSS, Tailwind CSS, Git, Docker, CI/CD
Concepts: Machine Learning, Deep Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Vector Embeddings, REST API, Microservices, Asynchronous Processing, Distributed Systems, Full-Stack Development

Work Experience

Ultipa USA Jun. 2025 – Sep. 2025
Software Engineer Intern

- Validated over 150 Graph Query Language (GQL) queries for schema creation, data insertion, and indexing, improving query accuracy and system stability by 25%.
- Developed a full-stack secure banking web application using React, Vite, and Tailwind CSS for the frontend and Python/Flask for the backend, integrating RESTful APIs and validating database operations to ensure 99.9% data integrity.
- Diagnosed and resolved data modeling and query execution defects; authored 10+ internal technical tutorials and improved documentation, reducing new developer onboarding time by 30%.

Projects

RAG Question Answering System | *Python, FastAPI, LangChain, LangGraph, OpenAI, Streamlit, Docker* | GitHub

- Built an end-to-end Retrieval-Augmented Generation (RAG) system with a FastAPI backend and Streamlit frontend, enabling context-aware question answering over user-uploaded PDF documents.
- Designed a LangGraph retrieval pipeline using OpenAI embeddings and a vector store with text chunking to improve retrieval precision across multi-document knowledge bases.
- Supported real-time PDF ingestion via a RESTful API, dynamically extending the knowledge base without service restart.

Translator Platform | *FastAPI, Celery, Redis, PostgreSQL, Hugging Face, PyTorch* | GitHub

- Designed and deployed a production-ready multilingual machine translation platform using FastAPI, Celery, Redis, and PostgreSQL.
- Built synchronous and asynchronous RESTful translation APIs with job persistence, real-time status tracking, and batch inference support.
- Implemented Redis-based response caching and API rate limiting, significantly reducing redundant model inference cost.
- Integrated Hugging Face Seq2Seq transformer models with dynamic language-pair routing for flexible multi-language support.

Banking Simulation App | *Flask, SQLAlchemy, SQLite, React, Vite, Tailwind CSS* | GitHub

- Built a full-stack banking system with a RESTful API backend supporting customer management, account operations, fund transfers, and merchant payments.
- Modeled relational entities (Customer, Account, Merchant, Transaction) with encapsulated transfer/payment flows, balance validation, and transactional consistency.
- Developed a responsive React frontend with account selector, transfer and payment forms, transaction history view, and multi-customer switcher.